Project Overview

Title: Heatwave Prediction Using Machine Learning
Course: ITAI 1371 – Final Project

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Overview:

- Predicting **heatwave occurrence (0/1)** using meteorological data (2006–2025)
- Classification problem
- Dataset highly imbalanced (I cant figure out how to put that one into my github folder.)
- SMOTE applied only to training
- Feature engineering added to improve model performance

Features Used:

- Temperature (TMAX, TMIN, TEMP2M)
- Wind, humidity, radiation, soil moisture
- Engineered: TEMP_RANGE, TEMP_AVG, SEASON indicators
- District one-hot encoding

Modeling Approach

Preprocessing:

- Train/Validation/Test split
- SMOTE on training only
- District features passed through

Models Trained:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- KNN
- SVC

Ensembles:

- Voting Ensemble (best 3 models)
- Bayesian Ensemble (GaussianNB)

Results & Comparison

Top Models:

- **Random Forest (Best Overall)**
  - Accuracy: **0.999772**
  - Precision: **1.000000**
  - Recall: **0.999544**
  - F1: **0.999772**
  - ROC-AUC: **1.000000**
- **Voting Ensemble**
  - Accuracy: 0.998630
  - ROC-AUC: 1.000000

Worst Model:

- **KNN**
  - Accuracy: 0.966446
  - Precision: 0.937126
  - Struggles with high-dimensional meteorological data

Bayesian Ensemble:

- Lower performance due to feature independence assumptions

Conclusion

Key Takeaways:

- Random Forest is the most reliable model for predicting heatwaves
- Ensemble methods (Voting) further stabilize predictions
- Feature engineering significantly improved accuracy
- SMOTE balancing was essential due to severe class imbalance